

the cost of downtime increases. We cannot generalise, however. As pointed out earlier, the load and the application determine the power supply requirements.

I think an example will throw more light on this: We cannot use a conventional UPS for process applications where we need single-phase output to support DCS loads with very stringent technical specifications like extreme temperature, vibration, and exposure to dirt, dust, corrosive gases and hazardous substances.

Mission-critical operations such as healthcare, broadcast and data centres require system performance with no interruption in service as the downtime can prove costly. What kinds of power backup systems are appropriate for them?

For mission-critical applications, the solutions are provided based on the uptime desired. The solutions are generally provided on the N+1 redundancy architecture.

You need an infrastructure that can anticipate and adapt to change. Emerson has pioneered an adaptive architecture for business-critical systems. Our dual-bus architecture for data centre applications ensures a fault-tolerant design with redundancy up to the load point. It essentially meets tier-4 level of criticality.

Which is better, flywheel-based UPS or double-conversion UPS with batteries?

There is no such thing as the perfect UPS system. It is the application and lifecycle cost with ease of operation and maintenance that decides the winner.

Static IGBT systems can output perfect sinewave voltage above on the linear load and with any distorted load-current waveform. These require less investment cost and are small, quiet and cheap to maintain. On the other hand, rotary systems permit a high output fault current without mains for downstream fault clearance. This means when the input utility power is off, rotary UPS systems can deliver more current during short circuit at the load end, enabling the circuit breaker to clear fault by tripping the faulty load.

Both static and rotary systems ensure 100 per cent availability of conditioned power through dual-bus architecture; their efficiency is comparable.

For western countries, where power disruptions are minimal and momentary (lasting only seconds), flywheel-based technology is useful. But for countries like India, where power disruptions are more frequent and last long hours, battery-backed UPS systems are preferable due to

their low capital expenditure, low operating expenditure, ease of service and easy-to-handle operations.

Considering batteries' unpredictable end of life, what new alternatives are being searched for?

Fuel cells are considered as more viable, environment-friendly and cost-effective technology in the long run. However, the high cost of fuel cells will continue to be a hurdle in the near future, though efforts are being made to lower their cost. Fuel cells for power systems are still emerging and haven't matured. So I don't foresee their wide usage in the near future owing to the demand-supply gap and high cost.

Fuel cells are similar to batteries—a chemical reaction

liberates electrons, which flow through an external circuit. However, in batteries, the reactants are fixed. Once the reactants exhaust, the battery dies. In fuel cells, the reactants are continually replenished. As long as the fuel and oxidant are flowing, the fuel cell produces power.

Another alternative is solar energy. It is estimated that the high cost of solar energy will be

an issue until 2020. The future trend of solar power depends on the national policy. Currently, solar power is reasonable only for special applications like rural, outdoor and demonstration.

What is the software-based approach to UPS scalability? What benefits does it bring to enterprises?

This question has two parts: soft scalability and intelligent power optimiser.

The software-based approach for soft scalability means customers can opt for capacity enhancement of their power system as and when the load demands. The hardware has to be essentially in place in the initial instance and the manufacturers offer this feature at an incremental cost. The customer benefit owing to this feature is questionable.

The interesting part is the latter one—intelligent power optimiser. Customers can gauge the load trend on the UPS, analyse the backup levels desired, study the downtime trend and plan capacity utilisation.

What to expect in future?

The technology of the future will be environment-friendly and have higher PF, higher efficiency, higher return on investment and lower total cost of ownership. We are still in the growth phase of the product lifecycle. Static UPS, especially double-conversion, will continue to do well in the Indian market. ●

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